Question Bank

Chapter 13: Earth as a System: Energy, Matter, and Life
Science, Grade 9 (Exploration) | With Answers

Sections: A. MCQs B. Fill in the Blanks C. Matching D. Assertion-Reason E. True/False F. Short Answer G. Long Answer

A. Multiple Choice Questions (1 mark each)

Q1. Which sphere consists of solid rocks, soil, landforms and the Earth's interior?

(a) Hydrosphere
(b) Geosphere
(c) Cryosphere
(d) Biosphere

Answer: (b) Geosphere

Q2. About 99 per cent of the Sun's energy reaching the Earth lies in which range of wavelengths?

(a) Gamma and X-rays
(b) Microwaves and radio waves
(c) Ultraviolet, visible and infrared
(d) X-rays and visible

Answer: (c) Ultraviolet, visible and infrared

Q3. The approximate value of the solar constant is:

(a) 1.4 kW m^{-2}
(b) 1 kW m^{-2}
(c) 3 x 10^{8} W m^{-2}
(d) 14 kW m^{-2}

Answer: (a) 1.4 kW m^{-2} (about 1400 J s^{-1} m^{-2})

Q4. The fraction of solar radiation reflected by a surface is called its:

(a) Insolation
(b) Albedo
(c) Solar constant
(d) Transpiration

Answer: (b) Albedo

Q5. Nearly all weather phenomena occur in which layer of the atmosphere?

(a) Stratosphere
(b) Mesosphere
(c) Troposphere
(d) Exosphere

Answer: (c) Troposphere

Q6. The ozone layer, which absorbs harmful UV rays, is located in the:

(a) Troposphere
(b) Stratosphere
(c) Thermosphere
(d) Exosphere

Answer: (b) Stratosphere (12 to 50 km)

Q7. During the day, the air flowing up the mountain slopes from the valley is called a:

(a) Mountain breeze
(b) Sea breeze
(c) Valley breeze
(d) Land breeze

Answer: (c) Valley breeze

Q8. Planetary winds are deflected towards the right in the Northern Hemisphere because of:

(a) Gravity of the Moon
(b) Rotation of the Earth
(c) Salinity of oceans
(d) Presence of clouds

Answer: (b) Rotation of the Earth on its axis

Q9. Large circular ocean current patterns are called:

(a) Gyres
(b) Drifts
(c) Monsoons
(d) Breezes

Answer: (a) Gyres

Q10. Which bacterium is found in the root nodules of legumes and fixes atmospheric nitrogen?

(a) Nitrobacter
(b) Pseudomonas
(c) Nitrosomonas
(d) *Rhizobium*

Answer: (d) *Rhizobium*

Q11. The conversion of nitrates back into nitrogen gas is called:

(a) Nitrification
(b) Denitrification
(c) Ammonification
(d) Assimilation

Answer: (b) Denitrification

Q12. Excess nitrates from fertilisers causing algal blooms in lakes is known as:

(a) Eutrophication
(b) Photosynthesis
(c) Transpiration
(d) Nitrification

Answer: (a) Eutrophication

Q13. Which international agreement has helped the ozone layer to slowly recover?

(a) Kyoto Protocol
(b) Paris Agreement
(c) Montreal Protocol
(d) Mission LiFE

Answer: (c) Montreal Protocol

Q14. Choose the most appropriate role of biogeochemical cycles in an ecosystem.

(a) To provide food directly to all organisms
(b) To recycle essential nutrients between biotic and abiotic components
(c) To create new elements for living things
(d) To remove pollutants from organisms

Answer: (b) To recycle essential nutrients between biotic and abiotic components

B. Fill in the Blanks (1 mark each)

Q15. The speed of electromagnetic waves in vacuum is _____ m s^{-1}.

Answer: 3 x 10^8

Q16. The amount of the Sun's radiation that reaches the Earth's surface is called _____.

Answer: insolation

Q17. In the troposphere, temperature decreases with height at about _____ degrees C per km.

Answer: 6.5

Q18. Near the equator, intense heating forms an equatorial _____ pressure belt, while sinking air at 30 degrees N and S forms sub-tropical _____ pressure belts.

Answer: low; high

Q19. Water with higher salinity is denser and therefore _____ to deeper levels of the ocean.

Answer: sinks

Q20. *Nitrosomonas* converts ammonia into _____, and *Nitrobacter* converts it further into _____.

Answer: nitrite; nitrate

Q21. Atmospheric CO_2 has risen by about _____ per cent since 1960 (from 315 ppm to 420 ppm).

Answer: 35

Q22. Most of the nitrogen used in fertilisers is fixed artificially by the _____ process.

Answer: Haber-Bosch

Q23. India's first insolation atlas, *Solar Radiation Over India*, was published by Anna Mani and _____.

Answer: S. Rangarajan

Q24. Clearing forests reduces photosynthesis and _____, which can lead to a decline in local rainfall.

Answer: transpiration

C. Match the Following (1 mark per pair)

Q25. Match the sphere with its example.

	Column A		Column B
1	Geosphere	A	Mangroves and coral reefs
2	Hydrosphere	B	Deccan plateau
3	Cryosphere	C	Air we breathe in forests and mountains
4	Atmosphere	D	Himalayan glaciers and polar ice caps
5	Biosphere	E	Ganga-Brahmaputra river system

Answer: 1-B (Deccan plateau); 2-E (Ganga-Brahmaputra river system); 3-D (Himalayan glaciers and polar ice); 4-C (air); 5-A (mangroves and coral reefs)

Q26. Match the organism with its role in the nitrogen cycle.

	Column A		Column B
1	*Rhizobium*	A	Converts nitrite to nitrate
2	*Nitrosomonas*	B	Return ammonia to soil (ammonification)
3	*Nitrobacter*	C	Converts nitrates back to nitrogen gas
4	*Pseudomonas*	D	Fixes nitrogen in root nodules of legumes
5	Decomposers	E	Converts ammonia to nitrite

Answer: 1-D; 2-E; 3-A; 4-C; 5-B

Q27. Match the person, institution or agreement with its contribution.

	Column A		Column B
1	Anna Mani	A	Recovery of the ozone layer
2	K.R. Ramanathan	B	Mapped solar insolation across India
3	IITM, Pune	C	Eco-friendly lifestyle initiative
4	Montreal Protocol	D	Computer models to simulate the Indian monsoon
5	Mission LiFE	E	Measured ozone levels in the Himalayas in 1934

Answer: 1-B; 2-E; 3-D; 4-A; 5-C

D. Assertion-Reason Questions (1 mark each)

Options: *(a) Both A and R are true and R is the correct explanation of A. (b) Both A and R are true but R is not the correct explanation of A. (c) A is true but R is false. (d) A is false but R is true.*

Q28. Assertion (A): Polar regions are much colder than equatorial regions.
Reason (R): Sun's radiation is spread over a larger area at the poles.

Answer: (a)

Q29. Assertion (A): Snow and ice keep polar regions very cold.
Reason (R): Snow and ice have low albedo and absorb most of the solar radiation.

Answer: (c) Snow and ice have *high* albedo and reflect a large proportion of radiation.

Q30. Assertion (A): Weather remains confined to the troposphere.
Reason (R): In the stratosphere temperature rises with height, so there is little vertical mixing of air.

Answer: (a)

Q31. Assertion (A): Venus is hotter than Mercury even though Mercury is closer to the Sun.
Reason (R): Venus has an atmosphere that has led to an uncontrolled greenhouse effect.

Answer: (a)

Q32. Assertion (A): Ozone at ground level is as protective for life as ozone in the stratosphere.
Reason (R): Ozone absorbs harmful ultraviolet radiation.

Answer: (d) Ground level ozone is harmful for health; only stratospheric ozone is protective.

Q33. Assertion (A): Plants and animals cannot directly use nitrogen gas from the air.
Reason (R): Nitrogen gas (N_2) is rather non-reactive.

Answer: (a)

Q34. Assertion (A): An excess of CO_2 in the atmosphere is undesirable although plants need it. Reason (R): Excess CO_2 intensifies the greenhouse effect and causes global warming.

Answer: (a)

E. True or False (1 mark each)

Q35. Like sound waves, electromagnetic waves need a medium to travel.

Answer: False. EM waves can travel through a vacuum; sound waves are mechanical waves and need a medium.

Q36. Dark surfaces have high albedo and stay cool.

Answer: False. Dark surfaces have low albedo; they absorb more radiation and heat up faster.

Q37. The height of the troposphere is maximum above the equator.

Answer: True.

Q38. A valley breeze blows down the slope into the valley after sunset.

Answer: False. That is the mountain breeze. The valley breeze flows up the slopes during the day.

Q39. Ocean currents help reduce temperature differences between the equator and the poles.

Answer: True.

Q40. The atmosphere holds about 71 per cent of the total global carbon.

Answer: False. About 71% is in the oceans; the atmosphere holds only about 1%.

Q41. Lightning contributes to the fixation of nitrogen.

Answer: True. It produces nitrogen oxides.

Q42. Gyres rotate counter-clockwise in the Northern Hemisphere.

Answer: False. They rotate clockwise in the Northern and counter-clockwise in the Southern Hemisphere.

Q43. Cities are usually warmer than surrounding rural areas (urban heat island effect).

Answer: True.

Q44. The Kyoto Protocol and Paris Agreement have been more successful than the Montreal Protocol.

Answer: False. The Montreal Protocol has been successful; the other two have been less successful.

F. Short Answer Questions (2 to 3 marks each)

Q45. What is albedo? Give two examples of surfaces with high and low albedo.

Answer: Albedo is the fraction of solar radiation reflected by a surface. Snow (0.80 to 0.90) and ice (0.50 to 0.70) have high albedo; black soil and ocean water have low albedo and are warmer.

Q46. Define the solar constant. Why is the insolation reaching the surface less than it?

Answer: The solar constant is the average solar energy received per unit time per unit area perpendicular to the Sun's rays at the top of the atmosphere, about 1.4 kW m^{-2}. Gases, clouds and dust absorb, scatter and reflect part of it, so under clear skies only about 1 kW m^{-2} reaches the surface.

Q47. Name the five spheres of the Earth system with one example of each.

Answer: Geosphere (Deccan plateau), hydrosphere (Ganga-Brahmaputra river system), cryosphere (Himalayan glaciers), atmosphere (the air we breathe), biosphere (mangroves, forests, coral reefs).

Q48. Why are equatorial regions warmer than polar regions?

Answer: The Earth is spherical, so the Sun's rays strike different latitudes at different angles. Near the equator radiation is concentrated over a smaller area, while at the poles it is spread over a larger area, giving less heating.

Q49. What is the urban heat island effect?

Answer: Cities are warmer than surrounding rural areas because buildings and roads of steel, concrete and asphalt absorb solar radiation and re-radiate heat, while rural areas have vegetation that cools through shade and transpiration.

Q50. How does the North Atlantic Drift influence climate?

Answer: It is a warm current, an extension of the Gulf Stream, that carries heat from the equator towards north-western Europe and keeps many ports ice-free in winter even at high latitudes.

Q51. Differentiate between the fast and the slow carbon cycle.

Answer: The fast cycle works over days to years: photosynthesis removes CO_2, and respiration and decomposition return it. The slow cycle works over millions of years: dead organisms are buried and form fossil fuels, which release carbon back as CO_2 when burnt.

Q52. What is eutrophication and what causes it?

Answer: Overuse of fertilisers adds excess nitrates to rivers and lakes, causing algal blooms. The algae deplete oxygen in the water and kill fish, threatening water bodies and coastal fisheries.

Q53. Why is the ozone layer important? Which chemicals damaged it?

Answer: It absorbs harmful UV radiation, protecting living organisms and ecosystems. Human-made chlorofluorocarbons (CFCs) from refrigerators and aerosols caused ozone loss over Antarctica.

Q54. How much solar energy falls on a 2 m^2 area in 2 hours if the insolation is 1 kW m^{-2}?

Answer: E = intensity x area x time = 1000 J s^{-1} m^{-2} x 2 m^2 x 7200 s = 1.44 x 10^7 J.

Q55. What is Mission LiFE?

Answer: Mission LiFE (Lifestyle for Environment) is an India-led global initiative introduced at the 2021 UN Climate Change Conference that encourages mindful, eco-friendly habits such as saving energy and conserving resources.

G. Long Answer Questions (5 marks each)

Q56. Explain how the Earth's atmosphere helps maintain a temperature suitable for life.

Answer: The atmosphere plays two crucial roles.
1. **It partly absorbs incoming solar radiation.** The ozone layer in the stratosphere absorbs harmful UV rays, and clouds and gases absorb some sunlight before it reaches the surface.
2. **It traps outgoing heat.** The Earth's surface absorbs sunlight and re-radiates it as infrared. Greenhouse gases such as CO_2, CH_4 and water vapour absorb part of this heat and keep the Earth warm.
Without the atmosphere the Earth would be too cold for life. However, excess CO_2 from human activities enhances the greenhouse effect and causes global warming. Venus, with its uncontrolled greenhouse effect, is hotter than Mercury.

Q57. Describe how uneven heating of the Earth gives rise to planetary winds.

Answer: Uneven heating between the equator and poles creates pressure belts.
- Near the equator, warm air rises, forming an equatorial low pressure belt. It moves poleward at height, cools, and sinks near 30 degrees N and S, forming sub-tropical high pressure belts. Surface air flows back to the equator.
- Part of the air at the sub-tropical belt moves poleward along the surface and rises near 60 degrees N and S, where it meets cold polar air, forming sub-polar low pressure belts.
- At the poles (90 degrees), cold dense air sinks, forming polar high pressure belts, and air flows towards the sub-polar belts.
The Earth's rotation deflects the winds to the right in the Northern Hemisphere and to the left in the Southern Hemisphere, so they follow curved paths.

Q58. Explain the processes of the nitrogen cycle. How would life be affected if nitrogen were not cycled?

Answer: Nitrogen gas (N_2) is non-reactive and cannot be used directly by most organisms.
1. **Nitrogen fixation:** *Rhizobium* (legume root nodules) and *Azotobacter* (soil) convert N_2 into ammonia; lightning also fixes nitrogen oxides.
2. **Nitrification:** *Nitrosomonas* converts ammonia to nitrite; *Nitrobacter* converts nitrite to nitrate.
3. **Assimilation:** plants absorb these compounds; animals obtain nitrogen by eating plants or other animals.
4. **Ammonification:** decomposers break down dead matter and waste, returning ammonia to the soil.
5. **Denitrification:** bacteria such as *Pseudomonas* convert nitrates back into N_2.
Without cycling, usable nitrogen would run out, so organisms could not make proteins and nucleic acids and life would not be sustained.

Q59. Describe the carbon cycle and explain how human activities have disturbed it.

Answer: Carbon moves between the atmosphere (CO_2), biosphere, geosphere (carbonate rocks, fossil fuels) and hydrosphere.
Fast cycle (days to years): plants take in CO_2 during photosynthesis; respiration and decomposition return it to the air.
Slow cycle (millions of years): buried dead organisms form coal, oil and gas; burning them releases CO_2 quickly. Oceans absorb CO_2 as carbonate and bicarbonate ions, which phytoplankton and shell-forming organisms use.
Human impact: burning fossil fuels and deforestation have raised atmospheric CO_2 by about 35% since 1960. This intensifies the greenhouse effect, causing global warming, melting of glaciers and sea ice, sea-level rise, extreme weather and more acidic seawater that threatens plankton and coral reefs.

Q60. Describe the interrelationship between the spheres of the Earth with examples.

Answer: The geosphere, hydrosphere, cryosphere, atmosphere and biosphere are connected by flows of energy and matter, so a disturbance in one affects the others.
Example 1: Less winter snowfall (cryosphere) means less water in a lake in summer (hydrosphere) and less grass growth for grazing animals (biosphere).
Example 2: Warmer Arabian Sea water increases evaporation and disturbs the southwest monsoon, causing floods in some regions and droughts in others.
Example 3: A warmer atmosphere accelerates melting of glaciers and polar ice, raising sea levels and threatening coastal cities and causing habitat loss.

Q61. What are the effects of deforestation on the Earth's spheres?

Answer: Deforestation affects air, water, land and living organisms:
- Less photosynthesis, so less CO_2 is removed and less O_2 is released; the carbon and oxygen cycles are disturbed.
- Reduced transpiration can decrease local rainfall.
- Surface albedo changes.
- Without roots to hold the soil, soil erosion increases.
- Habitats are destroyed, leading to loss of biodiversity.

Q62. Explain how winds blow as valley and mountain breezes, and how they help hilly regions.

Answer: Valley breeze (day): mountain slopes facing the Sun heat up faster than the valley floor. Air over the slopes becomes warm and rises, creating low pressure, and cooler air from the valley moves up the slopes.
Mountain breeze (night): slopes lose heat faster and cool; the cooler, denser air flows down into the warmer valley.
These local winds, commonly felt in places like Shimla and Dehradun, influence weather, agriculture and daily life, and help regulate temperature and moisture and support soil and crop health.